Brief Communication

Enlarged hands and feet – Not always acromegaly

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ABSTRACT

Pachydermoperiostosis maybe mistaken for acromegaly as it can present with progressive enlargement of hands and feet. We describe a 32 year old male with enlargement of hands and feet and extensive keloid formation. Family history was positive for similar complaints. X ray imaging showed normal heel pad thickness with acroosteolysis and subperiosteal new bone formation in hands and feet.IGF-1 was normal and glucose suppressed GH values were normal.

Key words: Pachydermoperiostosis, acromegaly, acro-osteolysis

Introduction

Pachydermoperiostosis or hypertrophic osteoarthropathy (primary or idiopathic), is an autosomal dominant disorder characterized by periosteal new bone formation that involves the distal extremities. The lesions present as clubbing of the digits and hyperhidrosis and thickening of the skin, primarily of the face and forehead. The changes usually appear during adolescence, progress over the next decade, and then become quiescent. During the active phase, progressive enlargement of the hands and feet produces a paw like appearance, which may be mistaken for acromegaly.[1]

Aims and objective

To describe a case of pachydermoperiostosis masquerading as acromegaly.

Materials and Methods

A 32-year-old male presented with enlargement of hands and forearms since the age of 18 years [Figure 1]. He also noticed enlargement of his feet and toes along with coarsening of facial features at around the same time. The enlargement was insidious in onset, uniformly progressive gradually over the years, until now. He also had a history of tendency towards keloid formation. He first noticed development of a keloid 4 years back after piercing his right ear followed by keloid development over the chest and hands after trivial trauma [Figure 2].

There was history of excessive sweating without any history of headache, visual field disturbances, or symptoms of obstructive sleep apnea. There was no history suggestive of another pituitary hormone excess or deficiency. A family history of similar complaints in younger brother and father

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Figure 1: Showing enlargement of hands with clubbing

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was present.

On examination, his height was 170 cm (50th-75th percentile), and weight was 85 kg with a body mass index of 29.4. He was hypertensive with blood pressure of 150/90 mm Hg. He had coarse facial features and oily skin without macroglossia, prognathism, or frontal bossing. He had severe clubbing of all fingers and toes with thickened distal extremities. Multiple keloids were noted. Systemic examination was normal.

Radiological evaluation showed a normal heel pad thickness with acro-osteolysis, subperiosteal new bone formation in hands and feet [Figures 3 and 4]. Routine investigations, including a complete blood count, blood sugar, serum calcium, phosphorous, and alkaline phosphatase levels were normal. Serum insulin-like growth factor 1 (IGF1) level was 150 ng/ml (115-307 ng/ml) and serum growth hormone (GH) levels after 75 g oral glucose was <1 μg/dl. Serum prolactin, cortisol, and thyroid hormone levels were normal.

Results

Based on presence of clubbing with classical radiological findings and after exclusion of GH excess, a diagnosis of pachydermoperiostosis was done. The keloids were excised.

Discussion

Pachydermoperiostosis was first described in 1868 by Friedrich of in two young brothers. In 1935, Touraine, Solente, and Gole applying an individual approach, described the pachydermoperiostosis as the primary form of hypertrophic osteoarthropathy, different from the majority of well-known secondary hypertrophic osteoarthropathy, which are always associated with the main cause (lung and heart). Ninety cases have been reported from 1947 to 1990.

An acromegalic phenotype in late childhood or early adulthood is shared by a variety of clinical conditions, including GH excess. Exclusion of an abnormality of the somatotrophic axis in a young patient with acromegaloid features is essential before considering other differential diagnosis such as pachy-dermoperiostosis or insulin mediated pseudoacromegaly, a disorder associated with severe insulin resistance.[2] In the absence of insulin resistance, findings characteristic of pachydermoperiostosis, such as thickening of the periosteum or the skin, clubbing, acrolysis, or alopecia should be looked for. In this case acromegaly was ruled out by a normal IGF1 and GH level less than 1 μg/ml, one hour after administration of 75 g of glucose. Insulin resistance was ruled out by a normal fasting and post prandial blood sugar, HbA1c and a normal fasting serum insulin levels of 11.1 MU/L (1.7-31 MU/L).



Figure 2: Extensive keloid formation on the anterior chest wall



Figure 3: X ray foot lateral view showing normal heel pad thickness



Figure 4: X ray left hand showing acral osteolysis and subperiosteal bone formation

Pachydermoperiostosis is inherited as an autosomal

dominant trait with variable expression. One-third of these patients have a positive family history[3] as was seen in this case. The pattern of inheritance in this case was suggestive of autosomal dominant transmission.

The onset is usually in adolescence, and the changes usually become progressively severe for 5-20 years and usually remain unchanged for life. A similar pattern of onset and progression were seen in this case.

The clinical features of pachydermoperiostosis are variable and have been classified as the complete form (pachydermia, clubbing, and periostosis); the fruste form (prominent pachydermia with minimal skeletal changes); and an incomplete form which has no pachydermia.[4] Our case had gross thickening of skin over face and scalp. He also had severe clubbing of all fingers and toes and had radiological evidence of periostosis. Thus, in our opinion, this patient had the complete form of pachydermoperiostosis.

Arthritis and joint effusions have been noted in 20-40% of cases but was not seen in our case.[5] Acro-osteolysis, a common radiological finding, affecting terminal phalanges of fingers and toes was seen in our case. Complications of pachydermoperiostosis include osteonecrosis of the femoral head, carpal and tarsal tunnel syndrome, and neurological changes due to compression of spinal cord and nerve roots. However, none of these complications were noted in our case.

Current treatment modalities for pachydermoperiostosis are limited. In our patient, plastic surgery consultation was sought, and the keloids were excised for cosmetic reasons.

Conclusion

In any case of enlarged hands and feet, we need to have an observant eye and keep in mind the possibility of pachydermoperiostosis, in differential diagnosis of acromegaly. This is important in view of the difference in treatment modalities of the two diseases.

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